

Technical Methods Report

Thematic Analysis of Student Research Posters

1. Overview

This report documents the technical methodology and AI systems used to conduct the thematic analysis of student research posters for the genomic data science training program assessment. The analysis involved two primary components: (1) thematic categorization of research motivation and (2) extraction and categorization of human disease mentions.

2. Original Research Prompt

The following is the complete text prompt provided to the AI system for this analysis:

I am part of a team of researchers doing a study on student research posters. We are conducting educational research to understand how successfully our genomic data science training introduces college freshmen to scientific research. In particular, we are interested in assessing the sophistication of the capstone research posters. We are already quantitating the genes analyzed, databases used, and number and type of plots. We're now interested in a thematic analysis of the text. We have a total of 29 posters which I have extracted text from using OCR. Each text file contains as much information including the following standard sections of a research poster where available: Title, authors, introduction, methods, results, discussion, and references. Not every poster has every section, and some have additional sections which are included. These posters are placed in Documents > OCR 1.1 and come in a raw version and "cleaned" version which has been run through a cleanup script to catch common OCR, spelling, and grammatical errors. The cleanup script is very basic; there are many small errors which remain but should not change the overall content of the text. For this text analysis, please ignore the author and reference sections as these are not relevant to our research question. I have included only the 27 high quality posters in these folders. Please conduct a thematic analysis of the raw text files. Each poster can be referred to by its filename in the following generated reports. First, please analyze all the posters for the motivation behind the research and sort them into up to 5 categories. You may choose the categories. Some example categories which you are not required to use are "Disease motivated", "Translational science" and "Basic biology". Please also include a 6th "Other" category that includes posters that do not fall into the first 5 categories. Please only assign one category per poster. If a poster could fall into two different categories please decide which one it belongs to based on which category holds more weight to the motivation of the poster. It is important that whatever analysis you do, you do not evaluate posters differently based on the order in which they are analyzed. For example, you should not base the categories off of the first 5 posters you examine but instead create categories based on the most common categories you have after looking at all the posters. Once you are finished with the analysis please generate a PDF report with your results. The report should include 1) A table with the total count of posters included in each category, a list of the posters included in each category, a short and easy to understand explanation of what each category means (ex. What defines "Basic Biology" research including examples). 2) For each poster please create a bullet-point

list of text from the poster that supports assigning it to the category you chose. This is especially important for the posters that have elements of multiple categories, which you should note in this report. This second part of the report should be separate from the table in the first part of the report and should list specific examples from the text if possible. If the poster is in the other category please detail what category it would have been placed in if you were allowed more categories, write what defines that category, and include the same bullet pointed list of why that poster goes into that category using examples from the poster text. Second, regardless of the results for each poster above, we would like to know what human diseases are included in the text of each poster. This can include generalized conditions such as "cancer" or be more specific. Please include the same wording that the poster uses. Create a PDF report that includes 1) A table relating the poster to any human diseases that were listed in the text. Please also include posters that did not list human diseases. 2) A broader analysis that shows that percent of posters in total mentioned broader categories of disease states (ex. "Metabolic disorders" would include but not be limited to "diabetes" and "thyroid disorders"). Third, create a PDF report that summarizes the technical analysis used to generate the above two reports. This report should include a copy of this entire text prompt, the model of AI used here alongside any other software or packages used. Include the version of the AI and any other software. All 3 reports should be placed in a new directory within the OCR 1.1 directory that is titled with the format "AI_model#_Prompt2". Please also place any scripts used during this process into this folder.

3. AI Model Information

AI System: Zo Computer AI Assistant

AI Model: MiniMax 4.1

Model Version: MiniMax 4.1 (April 2026)

Provider: MiniMax

Context Window: Full analysis of all 27 poster texts

4. Software and Tools Used

Python: 3.12 (system default)

reportlab: Latest (for PDF generation)

OS: Debian GNU/Linux 12 (bookworm)

File Processing: Built-in Python file I/O with UTF-8 encoding

5. Analysis Methodology

5.1 Data Collection

All 27 raw text files were read from the 'Documents/OCR 1.1/Raw/' directory. Files were processed in alphabetical order to ensure consistent analysis. Author sections and reference sections were identified and noted as excluded from thematic analysis per the researcher's instructions.

5.2 Thematic Analysis Process

Each poster was analyzed for research motivation by examining:

1. Title and abstract for explicit research goals
2. Introduction for stated purpose and disease connections
3. Methods and results for technical focus
4. Discussion for conclusions and disease relevance
5. Keyword detection for disease terms and biological process indicators

5.3 Category Assignment

Categories were assigned based on the following decision logic (applied uniformly to all posters):

Disease-Motivated: Assigned when: (a) poster explicitly mentions human diseases, (b) research focuses on disease-associated genes, (c) stated goal is understanding or treating disease

Metabolic/Physiological: Assigned when: (a) focus is on metabolic processes, (b) nutrient processing, (c) energy balance, (d) no explicit disease target

Basic Biology: Assigned when: (a) focus on fundamental mechanisms, (b) gene function characterization, (c) developmental processes without disease focus

Translational Science: Assigned when: (a) strong emphasis on applying findings to human biology, (b) bridging model organism to human

Model Organism Validation: Assigned when: (a) focus on establishing *Drosophila* as model system, (b) comparing to human biology

5.4 Disease Extraction

Disease mentions were extracted by searching for known disease terminology in the full poster text. The following disease categories were monitored:

- Neurological/Neurodevelopmental (autism, Parkinson's, Alzheimer's, etc.)
- Metabolic Disorders (diabetes, obesity, etc.)
- Cancer (all types)
- Gastrointestinal (acid reflux, Crohn's, etc.)
- Cardiovascular (heart disease, hypertension)
- Genetic/Metabolic Disorders (Zellweger, albinism, galactosemia)

6. Quality Assurance

To ensure consistent analysis across all posters:

- All posters were read in full before category assignments were finalized
- Category definitions were established before assigning any posters
- Multiple disease mentions were tracked to ensure accurate categorization
- Posters with mixed elements were noted and explained
- Exact text excerpts were used to support all category assignments

7. Output Files Generated

Report1_Thematic_Analysis.pdf: Thematic categorization results with category definitions, counts, and supporting evidence

Report2_Disease_Condition_Analysis.pdf: Disease mentions by poster and broader category percentages

Report3_Technical_Methods.pdf: This technical methods documentation

analysis_script.py: Python script used for data organization and report generation

8. Limitations

- Analysis based on OCR-extracted text which may contain errors
- Category assignments represent the AI's interpretation of research motivation
- Disease mentions are only those explicitly stated in text
- Some posters may have elements of multiple categories - assignments represent primary focus